ETALE COHOMOLOGY OF A DIRECTED LIMIT OF QUASI-COMPACT QUASI-SEPARATED SCHEMES IS THE COLIMIT OF THE COHOMOLOGIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: it supplies the representative complex, the pullback bookkeeping and the termwise colimit identity, and then cites the earlier variant lemma for the final identification.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter, so no load-bearing mathematics is replaced by a stub.

Editorial note. The label lemma-colimit-complexes occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Derived category D^+(S_etale) and the pullback of a complex to a scheme over S. Same chapter, the bounded below derived category of abelian sheaves on S_etale and the pullbacks K|_{X_i} and K|_X. Supplies the object K whose cohomology is being compared and the notation for its pullbacks.
- Representation of an object of D^+(S_etale) by a bounded below complex of abelian sheaves. Same chapter, the description of D^+(S_etale) by bounded below complexes. Supplies the complex G^bullet with G^n = 0 for n < a which the proof uses to replace K.
- Identification of the pulled back complex as the termwise colimit of the pulled back complexes. Same chapter, the pullback of a complex along the transition morphisms and the termwise colimit description F^bullet = colim f_i^{-1} F_i^bullet. Supplies the identity which, together with the cited variant lemma, gives the statement.

Source reference. Original lemma 0EZM: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7783–7783. Same-chapter navigation pointer invoked at proof line 7873, where the author cites the earlier variant lemma on colimits of complexes. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $X = \lim_{i \in I} X_i$ be a limit of a directed system of schemes over S with affine transition morphisms $f_{i'i} : X_{i'} \to X_i$. We assume that X_i is quasi-compact and quasi-separated for all $i \in I$. Let $K \in D^+(S_{étale})$. Then*

$$\operatorname{colim}_{i \in I} H^p_{étale}(X_i, K|_{X_i}) = H^p_{étale}(X, K|_X).$$

for all $p \in \mathbf{Z}$ where $K|_{X_i}$ and $K|_X$ are the pullbacks of K to X_i and X.

Proof. We may represent K by a bounded below complex $\mathcal{G}^\bullet$ of abelian sheaves on $S_{étale}$. Say $\mathcal{G}^n = 0$ for $n < a$. Denote $\mathcal{F}_i^\bullet$ and $\mathcal{F}^\bullet$ the pullbacks of this complex of X_i and X. These complexes represent the objects $K|_{X_i}$ and $K|_X$ and we have $\mathcal{F}^\bullet = \operatorname{colim} f_i^{-1}\mathcal{F}_i^\bullet$ termwise. Hence the lemma follows from Lemma 0EZM. □